

Highlights

SAFE-GTA: Semantics Augmentations-based Multi-Modal Fake News Detection via Global-Tokens Attention

Chaowei Zhang, Zongling Feng, Hui Sun, Jipeng Qiang, Yun Li

- The correlation between image-text can be better captured by global-tokens attention.
- QK sharing reduces parameter scales and ensures the efficacy of multi-modalities.
- Hierarchical semantics from BERT can be used to augment semantic representation.
- SAFE-GTA using unimodal LPMs avoids the restrictions of text-image alignment in fake news.

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ABSTRACT

Large pre-trained models (PLMs) have provided tremendous opportunities and potentialities for multi-modal fake news detection. However, existing multimodal fake news detection methods never manipulate the token-wise hierarchical semantics of news yielded from PLMs and extremely rely on contrastive learning yet ignore the different abstract levels between text and image. This paper proposes a novel multi-modal fake news detection method aiming at dealing with the issues mentioned above via 1) designing a Global-Tokens across-attention mechanism to capture the correlations between global-text and tokenwise-image representations (or tokenwise-text and global-image representations) obtained from *BERT* and *ViT*; 2) proposing a QK-sharing strategy embedded in the cross-attention to improve model efficiency and ensure the effectiveness of multimodal fusion; 3) deploying a semantic augmentation module by obtaining token-wise hierarchical semantic representation from the stacked blocks of *BERT* using *CNN* and *BiLSTM* to supplement the hierarchical textual representations for the task, thereby adjusting the unbalanced implications between modalities. We also prove the effectiveness of our approach by comparing it with four state-of-the-art baselines. All the comparisons are conducted on three widely adopted multi-modal fake news datasets. The results show that our approach outperforms the benchmarks by 0.8% in accuracy and 2.2% in F1-score on average across the three datasets.

1. Introduction

The controversy of fake news is a global topic that keeps being focused on by crowds since it was fermented by the 2016 USA presidential election and the COVID-19 pandemic Grinberg, Joseph, Friedland, Swire-Thompson and Lazer (2019); Patwa, Sharma, Pykl, Guptha, Kumari, Akhtar, Ekbali, Das and Chakraborty (2021) arose in the past few years. Recognizing the fake news that spreads on social media platforms is a big challenge because it is more attractive than real news for readers, which makes its propagation more uncontrollable Lazer, Baum, Benkler, Berinsky, Greenhill, Menczer, Metzger, Nyhan, Pennycook, Rothschild et al. (2018), thereby re-aggravating the propagation of fake news and then possibly arousing unpredictable negative consequences in perspectives of health, interests, economy, even social behaviors.

To against the negative effects mentioned above, there are numerous existing studies have been presented to deal with the task of fake news detection. Traditional related studies tackle the task by leveraging textual features (including content-based extractable latent features, such as sentiment, syntax, and style) Alonso, Vilares, Gómez-Rodríguez and Vilares (2021); Shu, Mahudeswaran, Wang, Lee and Liu (2020a), users' profile features Shu, Zhou, Wang, Zafarani and Liu (2019), social context features Nguyen, Sugiyama, Nakov and Kan (2020), and propagation network-based features Shu, Mahudeswaran, Wang and Liu (2020b) with the assistance of deep neural networks. The appearance of large pre-trained language models such as *BERT* Devlin, Chang, Lee and Toutanova (2018) and *T5* Raffel, Shazeer, Roberts, Lee, Narang, Matena, Zhou, Li and Liu (2020), has fundamentally revolutionized the landscape of fake news detection modeling. Such pre-trained language models enable grasping complex linguistic patterns, semantic nuances, and contextual cues present in language, thereby facilitating the ability to discern between reliable and misleading information by fine-tuning the pre-trained models Tian, Zhang, Wang and Liu (2020); Lin, Yi, Ma, Jiang, Luo, Shi and Liu (2023). For example, Rohit Kumar et al. proposed a *BERT*-based fake news detection model called *FakeBERT* by taking advantage of multiple parallel blocks of single-layered and kernel-size differentiated deep Convolutional Neural Networks Kaliyar, Goswami and Narang (2021).

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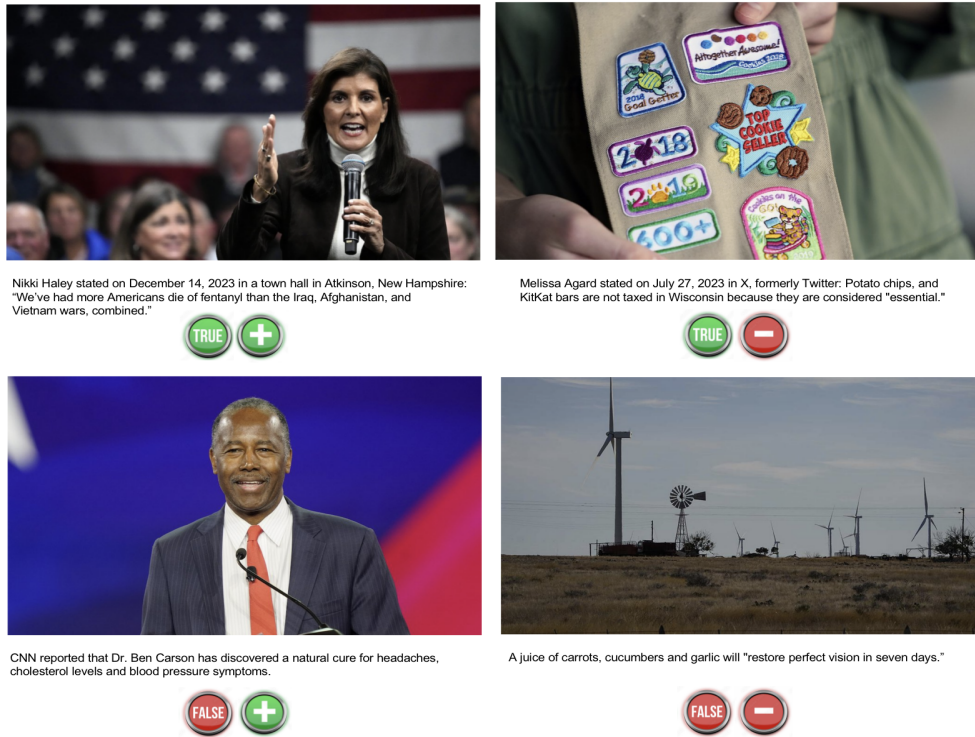


Figure 1: An example collected from PolitiCops that demonstrates four kinds of alignment relationships between text and image in fake and real news, where the True or False button in the figure respectively represents real or fake news, and the positive or negative button denotes whether a text-image pair is aligned or not.

News articles normally embrace plentiful text-based (such as title, content, etc.) and non-text-based (images, videos) information that allows in-field researchers to take advantage of such multimodal information for automatic fake news detection. Figure.1 demonstrates an example of four multimodal news from the *PolitiFact*, where the four samples denote the four kinds of relationships between text and image in fake and real news, which are true positive (i.e., aligned text and image in real news), true negative (i.e., unaligned text and image in real news), false positive (i.e., unaligned text and image in fake news), and false negative (i.e., aligned text and image in fake news). In contrast to text-based uni-modal fake news detection approaches, multi-modal fake news detection models leverage images that contain visual cues aiding in assessing the credibility of the news. Moreover, images can provide additional context to the textual information, which helps to identify discrepancies or inconsistencies in image-text pairs that might indicate potential misinformation or manipulation, thereby boosting the effectiveness of fake news detection Molina, Sundar, Le and Lee (2021).

In recent years, massive multimodal pre-trained models have been presented that have demonstrated powerful potential for multimodal representation-related downstream tasks Li, Selvaraju, Gotmare, Joty, Xiong and Hoi (2021); Radford, Kim, Hallacy, Ramesh, Goh, Agarwal, Sastry, Askell, Mishkin, Clark et al. (2021) Li, Li, Savarese and Hoi (2023). Such pre-trained models mainly utilize a text-image aligning approach called *contrastive learning*, which is implemented by using the *contrastive loss* to recognize the differences between *true-positive* image-text pairs and all of the other unmatched *true-negative* pairs (See Figure 1). This paradigm also can be transplanted to fake news detection Zhou, Yang, Ying, Qian and Zhang (2023); Yang, Zhu, Han and Hu (2023). However, the fake news detection systems adopting contrastive learning might yield several limitations: (1) the aligned text-image pairs in fake news act as *false positive* roles that reinforce the same false claims. Such alignment fosters a reinforcing effect where the images and texts mutually support the fake narrative, thereby increasing the difficulty for detection systems to identify fake news; (2) the image in a news normally is supplementary information of the text content, which means the semantic implications between image and text might have a huge gap due to varying levels of the abstraction in different modalities; (3) the text-image pairs originally obtained from different news may fall in a same semantic field (such

as the same/relevant topics or events) resulting in the pairs are inherently correlated, yet they are treated as negative instances by the existing contrastive learning methods. The effectiveness of contrastive learning approaches adopted in detecting multimodal fake news needs to be verified further, although the approaches have delivered a certain impact in the case.

In this study, we abandon utilizing the text-image aligning techniques and corresponding multimodal pre-trained models in our method, instead of adopting text- and image-based unimodal pre-trained models to avoid the calculation of semantic alignment between text and image. Specifically, our proposed semantic augmentation approach can make the model more concentrated on the information from texts, which allows the model to have better contextualized comprehensions in specific news(i.e., helps to deal with the limitations (1) and (2)). Meantime, the designed global-tokens cross-attention mechanism has an impressive ability to capture the consistency between image and text, which benefits handling the problem of semantic correlation between them(i.e., helps to deal with the limitation (3)). In this study, we propose a **S**emantics **A**ugmentations-based multi-modal **F**ake news detection approach using **G**lobal-**T**okens **A**ttention (a.k.a., **SAFE-GTA**), which leverages the document-level representations yielded from the stacked blocks of pre-trained language models(i.e., *BERT*) for semantic augmentation and embeds a global-tokens cross-attention mechanism for multimodal representation fusion.

The contributions of the study are listed as follows:

- We design a global-tokens cross-attention mechanism for multimodal representation fusion to effectively capture the correlations between text and image, in which the trainable Query and Key matrices for the two modalities are shared with each other.
- We manipulate token-level representations yielded from twelve stacked blocks of *BERT* using a Convolutional Neural Network for semantic augmentation.
- We propose **SAFE-GTA**, a multimodal fake news detection system implemented based on dual unimodal pre-trained models. Specifically, we adopt *BERT* for text learning, and *ViT* for image representation.
- We conduct a massive amount of experiments, and the results show that our approach significantly outperforms the state-of-the-art baselines by 0.8% in accuracy and 2.2% in F-score on average across three widely adopted datasets.

2. Literature Review

Identifying fake news has turned into a significant challenge in the current era of artificial intelligence. Existing fake news detection models oriented on inconsistent modalities of news data are fashioned on up-to-date algorithm architectures Vaswani, Shazeer, Parmar, Uszkoreit, Jones, Gomez, Kaiser and Polosukhin (2017) and large pre-trained models Devlin et al. (2018) Raffel et al. (2020) Radford et al. (2021) for classification decisions. This section systematically reviews the literature related to this study, in which the following two aspects are mainly inspected: unimodal-based fake news detection and multimodal fusion for fake news detection.

2.1. Unimodal Fake News Detection

The news published on social media contains rich multimodal data (a.k.a., images, texts, videos) that offer complementary insights, for prompting exploration into their combined contribution to the authenticity verification of news. Current unimodal approaches aiming at fake new detection primarily extract textual or visual features from news metadata for deep representation learning and higher-level unimodality interactions by leveraging neural networks and pre-trained models. Most of the mainstream studies focus on manipulating text Przybyła and Soto (2021); Bhatt, Goenka, Kalra and Sharma (2022) and images Xu, Sheng and Wang (2021) for fake news detection.

Text-oriented Approaches: Recent fake news detection research emphasizes the textual modality of news by analyzing textual contents Capuano, Fenza, Loia and Nota (2023); Kochkina, Hossain, Logan IV, Arana-Catania, Procter, Zubiaga, Singh, He and Liakata (2023), user profiles Jarrahi and Safari (2023), social metadata(i.e., retweets) Raza and Ding (2022); Allein, Moens and Perrotta (2023); Hamdi, Slimi, Bounhas and Slimani (2020), and the latent features implicated in the texts(i.e., sentiment, syntax) Lin, Kung and Leu (2022); Luvembe, Li, Li, Liu and Xu (2023); Guo, Zhang, Ding, Zhu and Yu (2023). For example, Qian and et al. propose a two-level(word and sentence) CNN-based generative model to capture new semantic patterns and detect fake news by analyzing users' historically significant responses Qian, Gong, Sharma and Liu (2018). Another study proposes a memory-guided multi-view fake

news detection method by considering the correlations among different domains. In this study, the authors leverage various textual features of news in perspectives of semantics, emotion, and style for feature fusion and representation aggregation Zhu, Sheng, Cao, Nan, Shu, Wu, Wang and Zhuang (2022). The existing text-oriented unimodal studies have demonstrated significantly expressive performance in fake news detection, yet the other modalities of news that can provide extra and rich learnable information for distinguishing fake news have never been involved. In this case, a better prospect can be imagined as soon as other modalities of news are integrated into text-only-based fake news detection systems.

Image-oriented Approaches: Image plays a crucial role in news reports, which can attract the attention of readers and provide the visible abstraction of news for audiences. In the field of fake news detection, images are normally used as the complementary information of text features, since the information implicated in them is limited for final decision-making. For example, Qi et al. give an insightful conception regarding news images that images consistently exhibit low resolution and powerful emotional content across the tasks Qi, Cao, Yang, Guo and Li (2019). To address these limitations, the authors propose a method by considering image features in terms of frequencies and pixel domains for shooting misinformation. Another study aiming to investigate the effectiveness of images extracts the propagation patterns from images to discover the discrimination between real and fake. Jin, Cao, Zhang, Zhou and Tian (2016).

In summary, unimodal-based approaches can effectively examine efficacious features of news, but these methods might ignore comprehensive understanding and crucial contextual cues present in other modalities. Thus, the robustness and accuracy of unimodal fake news detection systems could be enhanced by integrating other modalities.

2.2. Multimodal Fusion for Fake News Detection

Fake news can be manipulated through various means by malicious creators, including altering images and fabricating text content Xue, Wang, Tian, Li, Shi and Wei (2021), leading to un-optimal results being yielded if solely relying on single-modal approaches. To address this challenge, state-of-the-art studies aim to improve the performance of their systems by considering information from different sources by taking multi-modalities of news into account Jing, Wu, Sun, Fang and Zhang (2023) Comito, Caroprese and Zumpano (2023) Qi, Bu, Cao, Ji, Shui, Xiao, Wang and Chua (2023).

Intuitively, the feature representations of each modality can be separately or jointly obtained from large pre-trained models, such as *ResNet* Wu, Shen and Van Den Hengel (2019) and Vision Transformer(*ViT*) Dosovitskiy, Beyer, Kolesnikov, Weissenborn, Zhai, Unterthiner, Dehghani, Minderer, Heigold, Gelly et al. (2020) for image, *BERT* and *T5* for text, and CLIP for both image and text learning. An earlier study presents an Event Adversarial Neural Networks-based model, called EANN, which manipulates VGG-19 for image representation and TextCNN for word embedding acquirement. This study utilizes an auxiliary task of event classification to assist in feature extraction. Where the event classification branch aims to better disentangle the multimodal features, thereby enabling them to contain both specific event information and information independent of events. Another example, named HMCAN proposed by Qian et al. Qian, Wang, Hu, Fang and Xu (2021), is a hierarchical multimodal contextual semantics-based network implemented using ResNet and BERT for image and text representation learning, respectively. This model grasps different levels of semantics produced from layers of BERT, and the semantic representations are fused with image representation using three weight-shared multimodal attention blocks. In particular, the most recent works utilize the technique of contrastive learning that embeds an image-text alignment component constrained using contrastive loss Wang, Zhang, Xu, Xu, Xu and Wang (2023) Chen, Li, Zhang, Sui, Lv, Tun and Shang (2022). For example, Xu et al. propose a crossmodal contrastive learning framework for multimodal fake news detection, called COOLANT, which adopts a multimodal pre-trained model - CLIP for feature representation.

Multimodal fake news detection involves analyzing multiple types of data, such as text, images, and metadata, to identify misinformation. These approaches can significantly improve the performance of distinguishing fake news by obtaining fused information from various modalities. Overall, current multimodal pre-trained model-based methods deliver a visible and bright potential for fake news detection.

3. The Framework of SAFE-GTA

This section introduces our proposed SAFE-GTA model. SAFE-GTA aims to deal with the tasks of detecting fake news by utilizing the fusion information of cross-modalities and exploiting the different levels of semantic representations produced from the various blocks of pre-trained Text-Encoder. More specifically, the architecture of the SAFE-GTA model is composed of four components as shown in Figure 2:

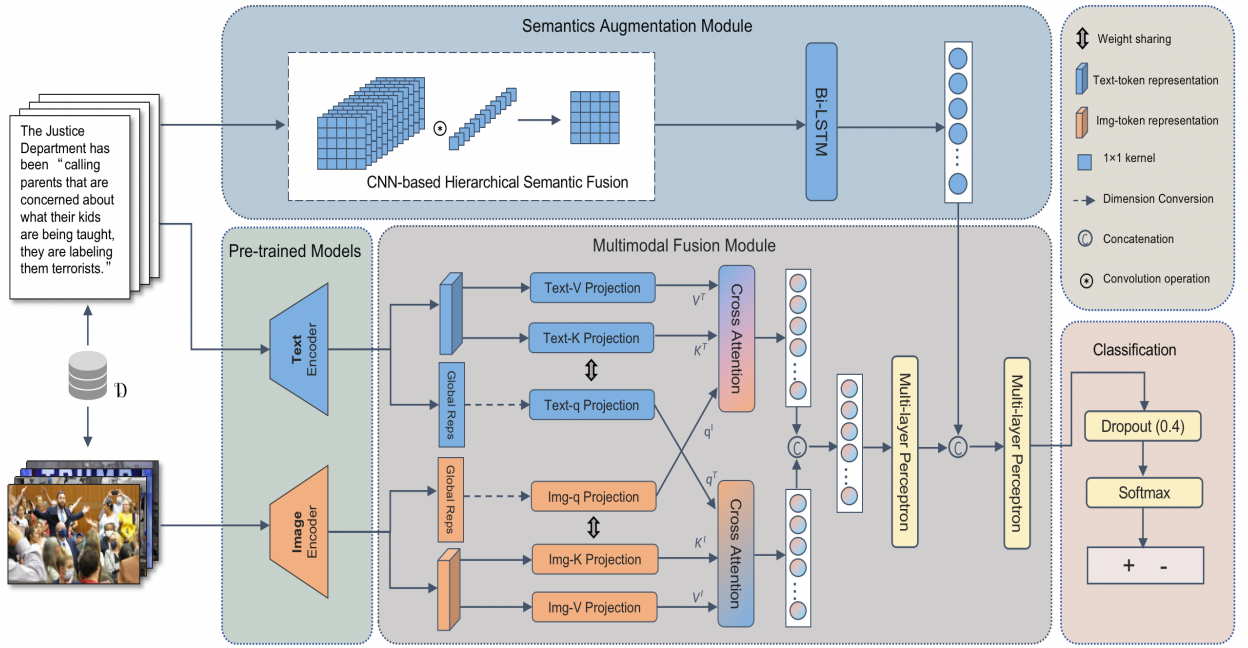


Figure 2: The framework of SAFE-GTA.

Multimodal encoding module. We employ BERT as our text encoder for textual representation extraction. To adapt the configuration of the BERT structure, we adopt ViT as the encoder for image learning. Such deployments make the encoders among modalities collaborative with each other for information contradistinction and fusion.

Multimodal fusion module. This module is designed to fuse the information between image and text, which adopts a weight-sharing strategy in terms of Query and Key matrices. The global and token representations produced from text and image encoders are used for cross-attention calculation by deploying the QK sharing and two Value projection layers. This module not only helps dynamically adjust the attention of the SAFE-GTA model on the global and token-wise information from image and text, but also can decrease the calculation cost, thereby improving the model's efficiency and the abilities of multimodal learning.

Semantic augmentation module. We obtain the text representation by presenting a semantic augmentation module, which fully utilizes the tokens' representations outputted from twelve stacked blocks of BERT. Moreover, the deployed convolutional kernels are used as the BERT-block-wise weights for semantic representation from shallow to deep during the period of convolution operations. Then, the learned augmented sequential representations will be sent to a Bi-LSTM network for generating augmented global semantic representation.

The binary classification module. The input of this module is obtained by concatenating the outputs yielded from the semantic augmentation module and the multimodal fusion module. Then, a multi-layer perceptron and softmax layer are deployed to predict the authenticity probability of news.

3.1. Image and Text Encoders

As we mentioned above, two unimodal pre-trained models - BERT and ViT -are deployed for text and image learning, and the global and token representations of text and image can be obtained from the pre-trained models.

Text representations Given a text sequence of news $T = \{t_1, t_2, \dots, t_n\}$, where n denotes the number of tokens in the text sequence. the sequence T is fed to BERT that can obtain (1) the token $[cls]$ corresponded vector as global representation e^{GT} ; (2) the embeddings of each token in the sequence $e^T = \{e_1^t, e_2^t, \dots, e_n^t\}$.

Image representations We utilize ViT, a pre-trained model upon the ImageNet dataset for image encoding. Each image is divided into 197 blocks(a.k.a, image tokens). Thus, an image sequence $I = \{i_1, i_2, \dots, i_{197}\}$ is conveyed to ViT for getting (1) the global representation e^{GI} that is the embedding corresponding to the token $[\#]$; (2) the token-wise

sequential representation $e^I = \{e_1^I, e_2^I, \dots, e_{197}^I\}$. Where e_1^I denotes the embedding of the image token i_1 after fine-tuning the pre-trained model.

The global representations - e^{GT} and e^{GI} stand for the overall comprehension of text and image, because the Bi-directional Transformer framework deployed in the unimodal encoders(i.e., Bert and ViT) allows each token in sequence to look over the correlations with all the other tokens using the self-attention mechanism.

3.2. Multimodal Information Fusion

The multimodal fusion mechanism in traditional multimodal fake news detection systems merely leverages the global representation of either image or text. In contrast, the token-wise sequential representation can retain the fine-grained features of the modality, which allows the capture of local features and structural information of a single modality. By combining the global representation and sequential representation with the cross-attention mechanism in the multimodal fake news detection task, the task model can adjust the focused range from global to local adaptively. This means that in the tasks demanding an emphasis on overall meaning, the task model is able to concentrate on comprehensive and global information. Conversely, for tasks that necessitate a focus on intricate details, the model is capable of shifting its attention to concentrate more intensely on a particular localized segment of the sequence.

The multimodal information fusion module not only takes advantage of the global and token-level representations captured by fine-tuning large pre-trained models but also adopts the strategy of Query(i.e., W^Q) and Key(i.e., W^K) weights sharing in the cross-attention mechanism during the period of information fusion. Specifically, we initialize two QK sharing layers for adapting the shapes of image and text representations(W^{TP} for text linear projection, and W^{IP} for image projection), and each modality goes through a value linear layer for obtaining value matrices(W^{TV} for text, and W^{IV} for image). The Q, K, and V representations of images and texts will be calculated via the four weighting matrices mentioned above. The following equations demonstrate the process of the calculations.

$$K^T = e^T \cdot W^{TP}, e^T \in R^{n \times h_t}, W^{TP} \in R^{h_t \times h} \quad (1)$$

where e^T denotes the embeddings of the tokens in sequence T , and W^{TP} is the QK sharing linear layer for text. The product of these two matrices - K^T indicates the key matrix of sequence T . The shape of K^T is in $R^{n \times h}$, where n is the length of sequence T , h is the outputted dimension of projection layer W^{TP} , and h_t is the dimensionality of text tokens' embeddings.

$$q^T = e^{GT} \cdot W^{TP}, e^{GT} \in R^{1 \times h_t} \quad (2)$$

where e^{GT} is the global representation of sequence T obtained by fine-tuning the BERT. q^T is the query vector of e^{GT} via the QK sharing weight matrix W^{TP} , which will be used to check its correlation to image sequence I .

$$V^T = e^T \cdot W^{TV}, W^{TV} \in R^{h_t \times h} \quad (3)$$

The QK sharing weighting matrix of images has a similar setting as that of text. The following equations show the process of producing the query vector q^I , key matrix K^I , and value matrix V^I of the image.

$$K^I = e^I \cdot W^{IP}, e^I \in R^{197 \times h_i}, W^{IP} \in R^{h_i \times h} \quad (4)$$

$$q^I = e^{GI} \cdot W^{IP}, e^{GI} \in R^{1 \times h_i} \quad (5)$$

$$V^I = e^I \cdot W^{IV}, W^{IV} \in R^{h_i \times h} \quad (6)$$

where the shapes of K^I , q^I are in $R^{197 \times h}$ and $R^{1 \times h}$, which are compatible with the dimensionalities of texts' q^T and K^T , respectively. The QK interactions between text and image have occurred in two employed cross-attention

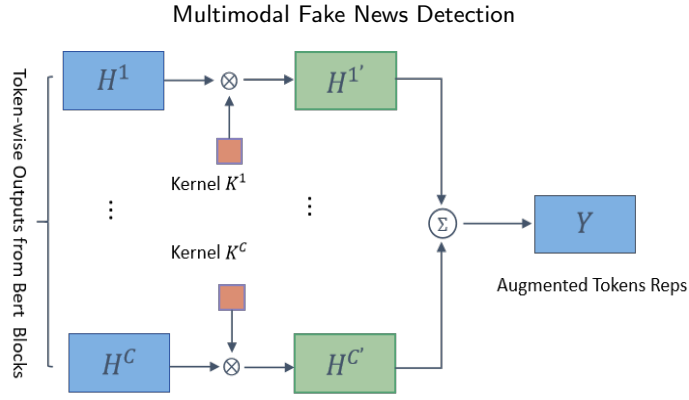


Figure 3: The insight of the CNN-based semantic augmentation module that can obtain the hierarchical semantic representation for text.

blocks as long as the query vectors and key matrices are obtained. Eq.7 and Eq.8 show the insights of global-token interactions in two cross-attention blocks between image and text.

$$h^{TC} = \text{Softmax}\left\{\frac{q^T(K^I)^T}{\sqrt{h}}\right\} \cdot V^T, \quad (7)$$

$$h^{IC} = \text{Softmax}\left\{\frac{q^I(K^T)^T}{\sqrt{h}}\right\} \cdot V^I, \quad (8)$$

where h^{TC} denotes the hidden representation of text that captures the correlation between the global information of text and the token-wise information of the paired image. In contrast, h^{IC} maintains the interactive information between images' global representation and texts' sequence information. Afterward, the two hidden representations h^{TC} and h^{IC} will be concatenated for information fusion as shown in Eq.9:

$$h^{TI} = \text{MLP}(\text{Concat}(h^{TC}, h^{IC})) \quad (9)$$

where MLP refers to a multi-layer perceptron that takes the concatenation of h^{TC} and h^{IC} as input for generating h^{TI} , which is the deep abstraction of fused text and image representation.

During the process of our preliminary experimental framework setup and the task model initialization, we found that the cross-attention with our deployed QK sharing strategy does not yield worse results compared to the traditional cross-attention mechanism. Moreover, the strategy cuts down the required number of parameter matrices in the cross-attention mechanism from six(double Q, K, V) to four(double shared QK, and V). This strategy not only ensures the effectiveness of the model but also reduces unnecessary additional parameter overhead, thereby improving the efficiency of the task model.

3.3. CNN-based Semantic Augmentation

A semantic augmentation module is designed in our model that fully takes advantage of the hierarchical semantics of text from BERT. Specifically, the proposed SAFE-GTA model fully exploits the different levels of token-wise representations(i.e., $\{H^1, \dots, H^C\}$) in a text sequence produced by the stacked blocks of BERT by deploying a CNN network, where the 1×1 convolutional kernels embedded in CNN is used to calculate the importance of the sequence representations in various levels of the blocks. Fig.3 illustrates the workflow of the CNN-based semantic augmentation module.

In the figure, the token-wise representations of 12 layers for a text sequence are treated as 3-dimensional data that embraces 12 channels. The values of corresponding convolutional kernels are randomly generated and trainable. Each channel is used for convolution calculation, and then the outputs from all the channels will be summed together as the

return of the semantic augmentation module, where the result maintains the hierarchical semantic information. It is notable that the training progress of this module is also constrained by the decision label. Eq.10 reveals the calculation process of hierarchical semantic representation for text.

$$Y = \sum_{c=1}^C K^c \otimes H^c, Y \in R^{n \times h_t} \quad (10)$$

where K^c represents a 1×1 kernel for channel c , and c is the index of channels from 1 to 12. H^c denotes to the tokens' hidden representations of an input text obtained from c th layer of BERT. The output Y is the token-wise summation of the results from 12 channels, which allows the module to dynamically learn and fuse the hierarchical semantics of texts orientating to decision-making. To obtain the whole representation of Y , we adopt a Bi-LSTM network to learn the sequential information of Y , which can be formalized as Eq.11.

$$z = BiLSTM(Y), z \in R^{1 \times h_t} \quad (11)$$

where z is the learned vector from Bi-LSTM by taking the Y as input. The shape of z is same as the dimensionality of word embeddings.

Overall, our proposed semantic augmentation module can: 1) flexibly capture the hierarchical semantics of text; 2) effectively strengthen the comprehension of textual information for the task model. Finally, the output of the module will be concatenated with the fusion representation produced from multimodal fusion module aiming to the classification between fake and real news.

3.4. Classification Module

In this module, we simply adopt a classic classification design that consists of a Dropout-enabled full-connection layer coupled with a Sigmoid function. This module takes the concatenation of outputs from the multimodal fusion module(i.e., h^{TI}) and the semantic augmentation module(i.e., z) to enrich the representation of news. Eq.12 delivers the process of computing the prediction of the SAFE-GTA model for a given image-text pair.

$$\hat{y} = Sigmoid(W_{fc} \cdot (h^{TI} \oplus z)) \quad (12)$$

where W_{fc} is the fully connected layer, and its shape is in $R^{2 \times (h+h_t)}$ for binary classification. Finally, the prediction $\hat{y}$ and the label y will be jointly sent to the loss function for model training. We adopt the following equation as our loss function as shown in Eq.13.

$$\mathcal{L} = \mathcal{L}_{CE} + \frac{\lambda}{2n} \sum_w ||w||^2 \quad (13)$$

where $\mathcal{L}_{CE}$ is the Cross-Entropy loss function, and $\frac{\lambda}{2n} \sum_w ||w||^2$ is a regular L2 regularization term that is used for improving the generalization ability of the task model.

4. Experimental Analysis

This section introduces the layout of experiment conduction to highlight the effectiveness of our proposed model. First, the configuration of our platform is described in detail. Second, the constitution of the datasets adopted to execute our experiments is clearly demonstrated. Third, we explain the process of the conducted contrastive experiments accompanied by corresponding results by referring to four popular and prestigious baselines. Finally, the ablation experiments are conducted to reveal the importance of the different components involved in our model.

Table 1

The statistics of the three datasets adopted for our model validation.

Datasets	Real News		Fake News	
	Training set	Test set	Training set	Test set
PolitiFact	647	75	1117	121
Gossipcop	6253	767	2135	166
ReCOVery	1105	123	194	22

4.1. Experiment Platform and Datasets

The platform used for executing our experiments is a high-performance computing server that consists of an Intel I9-3900 CPU - a 13th-generation processor with 24 cores and a base clock speed of 2.3 GHz, a 32GB of DDR5 RAM at a 5200 MHz speed, a 2TB M.2 2280 form factor NVMe-SSD offering high-speed data transfer and storage, and two NVIDIA RTX 3090 Turbo GPUs providing extraordinary graphical performance for deep learning applications. The learning rate is set to $3e - 5$, AdamW is used as the optimizer, and the batch size of data is equal to 16. We set up the experimental environment on the platform using PyTorch, which is the most popular Python-based open-source deep learning framework.

To verify the effectiveness of our model, we collect three widely-adopted real-world datasets to conduct experiments. The details of the datasets are shown in Table 1.

- *PolitiFact*¹ is a well-known non-profit fact-checking website that rates the accuracy of claims by officials, pundits, and other public figures. A tremendous number of in-field researchers collect data from the website aiming at unimodal or multimodal misinformation detection.
- *Gossipcop* is a resource specifically established for the detection and analysis of fake news, particularly in the context of entertainment news and celebrity gossip. This dataset is key for research and development in the field of misinformation studies, especially with a focus on multimodal data.
- *ReCOVery* is a specially designed and constructed repository aimed at supporting research efforts in the fight against misinformation related to COVID-19.

Table.1 illustrates the statistics of the three deployed datasets. Specifically, The *PolitiFact* dataset maintains 722 real news(i.e., 647 for training and 75 for testing) and 1238 fake news(i.e., 1117 for training and 121 for testing), the *Gossipcop* consists of 7020 real news(i.e., 6253 for training and 767 for testing) and 2301 fake news(i.e., 2135 for training and 166 for testing), and 1228 real news items(i.e., 1105 for training and 123 for testing) and 216 fake ones(i.e., 194 for training and 22 for testing) are contained in *ReCOVery* dataset. Although all three datasets are unbalanced, they are still adopted for conducting method verification in many studies.

4.2. Baselines and Evaluation Metrics

To spotlight the effectiveness of our approach, we employ four well-known and representative existing multimodal-based fake news detection benchmarks by collecting their code from GitHub² or replicating their experiments using *PyTorch* according to the implementation introduction from their paper.

Spotfake: Singhal, Shah, Chakraborty, Kumaraguru and Satoh (2019) employs pre-trained language models, such as BERT, for learning textual data, and uses VGG-19, which is pre-trained on the *ImageNet* dataset, to extract features from images.

Spotfake+: Singhal, Kabra, Sharma, Shah, Chakraborty and Kumaraguru (2020) strengthens SpotFake by utilizing XLNet to extract textual features. These multimodal features are then fed into a binary classifier as long as the text and image features are concatenated.

¹<https://www.politifact.com/>

²<https://www.github.com/>

Table 2

The comparison results between four baselines and the proposed SAFE-GTA model evaluated on PolitiFact, Gossipcop, and ReCOVery datasets.

Datasets	Methods	Accuracy	Real News			Fake News		
			Precision	Recall	F1-score	Precision	Recall	F1-score
PolitiFact	Spotfake	0.796	0.804	0.884	0.843	0.777	0.653	0.711
	Spotfake+	0.842	0.830	0.933	0.879	0.866	0.693	0.770
	HMCAN	0.847	0.902	0.843	0.871	0.772	0.853	0.810
	CAFE	0.882	0.865	0.958	0.909	0.919	0.76	0.832
	SAFE-GTA	0.888	0.896	0.925	0.916	0.873	0.826	0.849
Gossipcop	Spotfake	0.846	0.848	0.168	0.281	0.846	0.993	0.914
	Spotfake+	0.866	0.669	0.488	0.565	0.895	0.947	0.928
	HMCAN	0.860	0.650	0.469	0.54	0.891	0.945	0.917
	CAFE	0.871	0.819	0.356	0.495	0.875	0.983	0.926
	SAFE-GTA	0.876	0.704	0.531	0.605	0.903	0.951	0.927
ReCOVery	Spotfake	0.869	0.714	0.227	0.344	0.876	0.983	0.927
	Spotfake+	0.903	0.700	0.636	0.667	0.936	0.951	0.943
	HMCAN	0.924	0.923	0.545	0.687	0.924	0.991	0.957
	CAFE	0.903	0.900	0.409	0.563	0.903	0.991	0.945
	SAFE-GTA	0.937	0.933	0.636	0.756	0.938	0.991	0.964

HMCAN: Qian et al. (2021) establishes multimodal attention networks using self-attention mechanisms to combine textual and visual features. In the case of news that lacks images, HMCAN generates placeholder images to create text-image pairs.

CAFE: Chen et al. (2022) intelligently combines the individual features of two modalities and the extracted interaction representation between multi-modalities to identify fake news using contrastive learning.

Metrics: Accuracy is a frequently used evaluation metric in binary classification tasks including fake news detection. Nonetheless, its reliability diminishes significantly when dealing with imbalanced datasets. In our experiment, we have included additional evaluation metrics, namely Precision, Recall, and F1 score, to provide a more comprehensive assessment of the task's performance in such situations.

4.3. Effectiveness Evaluation

The conducted experiments in this study are in two folds: (1) the comparison with the referred four baselines mentioned above; and (2) the ablation study by disabling the multimodal fusion module or semantic augmentation module. Table.2 demonstrates the statistical results that involve the performance comparisons between five methods and our SAFE-GTA model on PolitiFact, Gossipcop, and ReCOVery datasets. From the table, it is observable that our model significantly outperforms the four baselines in most metrics. More detailed, we have the following observations and analysis that can spotlight the effectiveness, robustness, and superiority of our model.

Effectiveness on task classification: SAFE-GTA achieves the best performance in almost all metrics on three datasets. Specifically, (1) under the comparison on PolitiFact, SAFE-GTA has the highest accuracy (i.e., 0.888), indicating it is the most reliable method for the classification between fake and real news. Moreover, the values of SAFE-GTA's precision, recall, and F1-score are fairly high for individual fake or real news evaluation, which means SAFE-GTA is quite adept at correctly identifying real or fake news; (2) SAFE-GTA again shows strong performance with the highest accuracy (i.e., 0.876) and precision on fake news evaluation (i.e., 0.903) on Gossipcop dataset, which

outperforms the second best model - CAFE by 0.5% and 2.8%, respectively. It has moderate precision and recall for real news but excels with the highest F1-score for fake news (i.e., 0.927), which suggests a good balance between precision and recall in this category; (3) SAFE-GTA outperforms all other baselines in all metrics for the ReCOVery dataset, marking it as the superior method for detecting both real and fake news in this context. The high accuracy (i.e., 0.937), precision (i.e., 0.936 on average across fake and real), recall (i.e., 0.814 on average), and F1 score (i.e., 0.86 on average) indicate that SAFE-GTA is very effective at generalizing its detection capabilities in this dataset.

Robustness Across Datasets: SAFE-GTA consistently ranks as one of the top-performing methods across all three datasets, highlighting its robustness and adaptability to different types of news content. Meanwhile, SAFE-GTA maintains a high F1-score, especially on the Gossipcop and ReCOVery datasets while some baselines struggle to balance precision and recall for fake news. The robustness of our model benefits from the dual QK-shared global-to-token cross-attention in SAFE-GTA that is able to capture the relevance between image and text in perspectives of whole representation and token-wise sequence representations. In other words, our proposed global-to-token cross-attention helps the SAFE-GTA model possess a better ability to discover the correlations and inconsistency between modalities by a global viewpoint of text or image.

Superiority in Complex Environments: The datasets adopted for experiment conduction are inconsistent with each other in terms of topics, lengths of news, consistency between modalities, and other factors. However, SAFE-GTA can achieve a relatively outstanding performance across all the datasets compared with other baselines. Such superiority of SAFE-GTA is caused by the advantages of the embedded semantic augmentation module, which grasps different levels of token-wise representation for text, thereby boosting SAFE-GTA has better capability on hierarchical text information comprehension produced from the stacked blocks of BERT. It can be observed from Table 2 that SAFE-GTA monopolizes the best accuracy and F1 scores on both fake and real testing excepting the F1 score for fake news on the Gossipcop dataset (i.e., 92.7% that slightly drops 0.1% compared with 92.8% produced by Spotfake+). Thus, SAFE-GTA has the relatively best performance to precisely locate both real and fake news among the models. Meantime, it is suitable to transfer SAFE-GTA to news' veracity-related real-world applications, such as social media platforms and news aggregation services.

Summarization from the comparison result: SAFE-GTA is proven as a highly effective method for fake news detection by conducting the performance comparison with other outstanding benchmarks in this study. Meantime, the experimental results demonstrate the effectiveness, robustness, and huge potentiality of SAFE-GTA in dealing with misinformation in sensitive and rapidly evolving news environments.

4.4. The Ablation Study

To assess the impact of the embedded components in our approach, we conducted an ablation study by shutting down the interfaces of the fusion block and the semantic augmentation module. To achieve this goal, we design the following SAFE-GTA variants to examine the importance of such components.

- CF-GTA: represents the SAFE-GTA model that disables the multimodal information fusion module.
- SA-GTA: denotes another SAFE-GTA variant that closes the functionality of semantic augmentation.
- FA-GTA: indicates the SAFE-GTA that shuts down the effects of both information fusion and semantic augmentation.

Figure.4 clearly depicts the results yielded from SAFE-GTA and its variants in terms of accuracy, and F1-score for either real or fake news on three datasets. Also, it helps to understand the contributions of the components to the performance of the original SAFE-GTA. To better highlight the implications of the ablation study, we provide the following viewpoints by observing and analyzing the results from the figures.

GTA serves as the baseline with the highest performance across all metrics on three datasets compared with others. CF-GTA shows a drop in all metrics compared with SAFE-GTA, indicating the importance of the ablated fusion module in recognizing both real and fake news. SA-GTA illustrates a drastic decrease in both F1 scores, with a larger impact on F1-True, again highlighting the importance of the semantic augmentation module in classifying news. FA-GTA shows the relatively least variation across the metrics.

The CF-GTA variant's performance indicates that the multimodal fusion module is critical for the model's ability to identify fake news accurately across datasets, particularly in the ReCOVery dataset where it also affects the detection of fake news. On the other hand, The performance gaps between SAFE-GTA to SA-GTA, especially in the F1-True

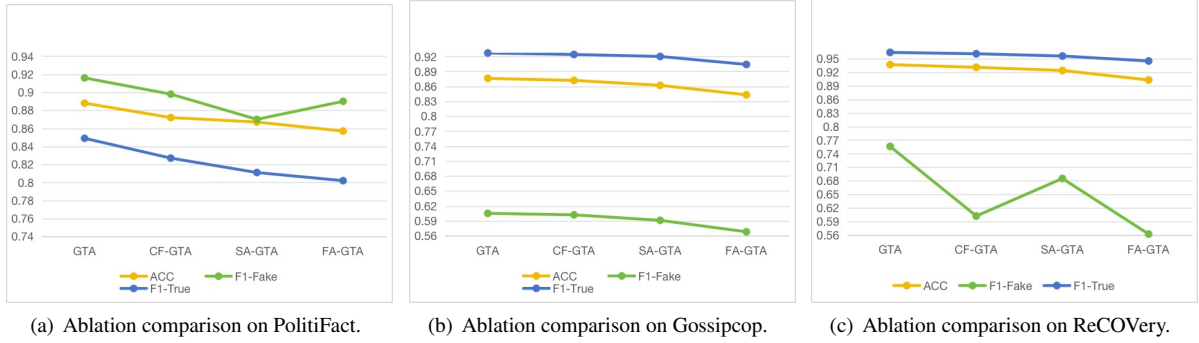


Figure 4: Ablation comparisons among SAFE-GTA variants in terms of accuracy, F1-score on Fake news, and F1-score on Real news on three datasets.

metric, highlight the importance of the semantic augmentation block. That means semantic augmentation in the model plays a nuanced role, which potentially offers a trade-off between accurately detecting real and fake news.

The varying impacts of different ablated components validated on three datasets suggest that the characteristics of each dataset, such as the scale of fake and real news, information alignment among modalities, and the length of news items significantly influence which model features are most important. In summary, the ablation study clearly demonstrates that the performance of the SAFE-GTA model in terms of various metrics is reliant on the integrated functioning of multiple components. Any single module obliteration tends to reduce the performance to varying degrees. The study also underscores the necessity for tailored strategies for different datasets and contexts when combating fake news.

5. Conclusion

This work proposes an efficacious semantic augmentation-based multi-modal fake news detection model by taking advantage of multimodal fusion and levels of fine-grained semantic information yielded by large pre-trained models. Specifically, we design a QK-shared dual global-token cross-attention mechanism that not only captures the interactive relationship between image and text in terms of global representation and tokens representations to benefit the information fusion crossing modalities but also decreases the number of parameters, thereby improving the efficiency and performance of our proposed SAFE-GTA. Moreover, we present a semantic augmentation module that adopts CNN to grasp different levels of token-wise semantic representation produced by twelve stacked blocks of BERT, and then the learned token representations that maintain the semantic information from deep to shallow will be transferred to a Bi-LSTM layer for obtaining vectorized representation. Finally, the outputs yielded by information fusion and semantic augmentation will be concatenated for decision-making. The experimental result reveals that our approach outperforms the baselines by 0.8% in accuracy and 2.2% in F1-score on average across three widely adopted datasets. In the future, we will further investigate the efficiency of multimodal fusion from the perspective of tokens-tokens cross attention, develop a more effective semantic augmentation technique by leveraging the contrastive similarity among multi-modalities, and other state-of-art large unimodal or multimodal pre-trained models will be considered adopting in the construction of task models.

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CRedit authorship contribution statement

Chaowei Zhang: Conceptualization of this study, writing - Original draft preparation. **Zongling Feng:** Data collection, and experiment implementation. **Hui Sun:** Data analysis and interpretation. **Jipeng Qiang:** Critical revision of the article. **Yun Li:** Final approval of the version to be published.

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